4	**(a)**	straight line through origin	M1	
		shows acceleration proportional to displacement	A1	
		negative gradient	M1	
		shows acceleration and displacement in opposite directions	A1	[4]

(b) **(i)**	2.8 cm	A1	[1]
(ii)	*either* gradient = ω^2 and $\omega = 2\pi f$ *or* $a = -\omega^2 x$ and $\omega = 2\pi f$	C1	
	gradient = 13.5 / (2.8 × 10^{-2}) = 482		
	ω = 22 rad s^{-1}	C1	
	frequency = (22/2π =) 3.5 Hz	A1	[3]
(c)	e.g. lower spring may not be extended e.g. upper spring may exceed limit of proportionality / elastic limit (*any sensible suggestion*)	B1	[1]

5 **(a)** **(i)** ratio of charge and potential (difference) / voltage